Landauer and Phase-Space Accounting

Why region reduction costs heat in principle — and why the content is irrelevant

FFGFT A-Series · A280

Standalone edition with layer markings, check-script references and a complete source list

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Abstract. The Landauer bound is developed here not as a statement about *information* but as a statement about *regions in phase space*. A physical system occupies a region; a map that reduces several source regions onto one target region shrinks that volume; Liouville's theorem forbids volume shrinkage in a closed system; hence a bath must absorb the volume, and the price is heat $Q \geq k_B T \ln(W_{\text{before}}/W_{\text{after}})$ — with $\ln 2$ merely the binary special case. Counting proceeds by distinguishability (orthogonality), not by energy; the absolute base discretisation is supplied by h^f. Content neutrality follows immediately: thermodynamics counts regions, it does not read them. Three levels are kept strictly apart — construction (engineer), state transition (clock), description (program) — and from this it follows that a real computer does not erase bits discretely but carries a continuous entropy current (DRAM refresh, qubit decoherence). The epistemic caveat is booked explicitly: not measurable does not mean not present — Landauer holds because of the mechanics (Liouville), not because of ignorance; the irreversibility is statistical, not ontologically absolute. *Contact with the ξ scheme of FFGFT: none* — Landauer accounting lives at the level of statistical mechanics (its own level, its own accounting).

Layer markings. **[STIPULATION]** declared starting point (German corpus token: [SETZUNG]) — **[B]** derived algebraically/mathematically from declared foundations — **[K]** empirically tested or documented from primary sources — **[S]** plausibility sketch — **[H]** open research question (an extension of the A-series scheme; see the appendix, item 6).

Check script. `landauer_pruefskript.py` (checks 1–10). Every numerical statement in this document is backed there by assertions. Check 7: state count in the colloid region; check 8: analogue case with declared δx.

1 The region comes first

Two glasses stand on a table — region A and region B. Next to them, the world ocean. One of the glasses is poured into the sea. The volume of the glass does not vanish — it distributes itself across 1.4 billion cubic kilometres. This image carries the whole document: there is no vanishing information, no annihilated bits — only a reduction of regions that has a measurable price.

[STIPULATION] A physical system occupies a region in phase space. That region has a size — measurable in units of h^f (Section 6). It exists independently of how one names, partitions or describes it. That is the physical starting point. Everything else comes afterwards.

[B] Descriptions come from outside. If one decides to partition the region into two subregions and to call them "A" and "B", and the target region "Z", one has introduced a description of the region reduction. If one calls them "0", "1" and "0" instead (the traditional notation), the confusion with logical bits arises — that convention hides the fact that "0" denotes both a source region and the target region, and it suggests an information content where there is none. If one partitions more finely — into 1000 zones — one has introduced ~ 10 logical bits. If one merely assigns a number between 0 and 1, one has introduced an analogue value. Physics enforces none of these decisions. In all cases the region is the same physical region.

The terms "bit", "byte", "qubit", "analogue value" are labels for descriptive decisions — not for physical objects. They partition the region; they do not create it.

1.1 The Landauer bit is hardware — not information

[B] This is regularly confused. The clarification in one paragraph: a Landauer bit is a physical region with two stable macrostates — a transistor, a capacitor, a trace held high or low. A bus bit is the same: a physical region with two stable macrostates, embedded in a parallel hardware architecture. A memory cell is the same. The three are structurally identical. Thermodynamics treats all three alike:

$$Q \geq k_B T \ln\left(\frac{W_{A+B}}{W_Z}\right)$$

— regardless of whether the region is called "Landauer bit", "bus bit" or "memory cell".

The hardware is fixed before any computation runs. It does not change because of what one computes with it. A 64-bit bus has 64 physical regions — before the computation, during it, after it. What the computation does with those regions — which maps it executes, which regions are reduced onto which target regions — is a logical description of those physical processes. The description does not exist in the hardware; it is an external interpretation.

The information content of a bit — whether it means "true" or "false", whether it belongs to a correct computational step, whether it is read at all — is a decision at the level of description. Thermodynamics does not know that level. It knows only: region A and region B exist; after the region reduction the target region Z exists; $W_Z \leq W_{A+B}$; the bath absorbs the difference.

1.2 Three levels — strictly separated

[B] Level 1 — construction. The engineer decides at chip-design time which region operations exist — bijective or non-bijective. That is cast in silicon, unchangeable during operation, prior to any computation. This decision fixes the thermodynamic cost structure — not the computation, not the program.

[B] Level 2 — state transition during operation. Once the computer runs, state transitions occur clock cycle by clock cycle — uniformly, at the clock frequency, independently of what is being computed. Landauer costs arise only for **non-bijective** region maps — when a component, by its physical construction, reduces several input regions onto one target region Z. That is fixed by the hardware architecture, not by a computational operation. A transistor flipping from region A to region B executes a bijective map — no volume loss, no Landauer cost. What the descriptive level subsequently does with the component's state is thermodynamically irrelevant.

Landauer costs run with the clock frequency, practically constant, as long as the computer is running — because the non-bijective region maps are fixed by the designer and are incurred with every clock cycle. CPU utilisation says little about this: it measures how many clock cycles are used for computations at the descriptive level — but at idle the clock transitions keep running. The thermodynamic difference between high and low utilisation is small as long as no physical parts are actually switched off. Clock gating, power gating — these are hardware decisions at level 1, not program decisions. The hardware determines the cost — not the program.

[B] Level 3 — description. The program, the computation, an AI controller may decide which hardware is used — switch off parts, choose routes, determine that for a given result the full bit width is not needed. This has nothing to do with Landauer. It is resource management at the descriptive level. The thermodynamic costs of the hardware in use are incurred regardless — fixed by construction and switching moment, independently of what the descriptive level does with the results.

1.3 How much is inside a physical region?

[K] A colloidal particle in a 1 µm trap physically contains roughly **7.9×10^9 orthogonal quantum states** *(check script, check 7)*. If one coarsely names two zones "left" and "right", one describes that region with 1 logical bit. Partitioned into 1000 zones, one describes the same region with ~ 10 logical bits. The physical region is identical in both cases — the depth of description is the observer's choice.

[B] **The cost of erasure attaches to the region, not to the description.** The formula reads

$$\boxed{Q \;\geq\; k_B T \ln\!\left(\frac{W_{\text{before}}}{W_{\text{after}}}\right)}$$

W_{before} and W_{after} are physical region sizes in units of h^f. How one names that ratio — "erasing 1 bit" or "reduction by a factor of 2" — changes nothing about the physics.

[K] **The analogue case shows this particularly clearly — and shows its complexity at the same time.** How many distinguishable states an analogue signal contains depends on several factors, all of them physical: on the bandwidth B, the temperature T and the resistance R (Johnson–Nyquist noise, $U_{\text{rms}} = \sqrt{4k_BTRB}$ [B20]), on the sensitivity of the measurement apparatus, on the signal-to-noise ratio of the concrete system, and on further external sources of interference. There is no universal resolution limit for analogue signals — it is a system property, not a constant of nature.

What can be said in general: if the resolution limit is set by the thermal noise of the bath itself (the physically deepest case), then the principle already stated in Section 6 applies: the same bath that collects the erasure cost limits the gradation of the analogue signal. The region count $W = L/\delta x$ and the erasure cost $k_BT \ln W$ scale with the same temperature quantity — the accounting is self-consistent. How many logical bits one ascribes to the analogue region is independent of this and remains a descriptive decision *(check script, check 8: illustrative example with declared δx)*.

1.4 Shannon and Boltzmann are different ledgers

[B] [B18][B19] Shannon describes how much uncertainty a *description* carries: $H = -\sum_i p_i \log_2 p_i$ — a quantity about descriptions. Boltzmann describes how large a *physical region* is: $S = k_B \ln W$ — a quantity about regions. The two are identical only if one maps the description exactly onto the physical microstates and those are uniformly distributed. In every other case — coarse partitioning, unequal occupation, continuous systems — they are different things with different units.

[K] **What Landauer actually said in 1961** [B1]: he wrote explicitly "logically irreversible" — logically, not physically irreversible. He meant a map in which different logical input states lead to the same logical output state. That is a statement about descriptions.

[B] **And therefore, taken by itself, it is not a physical condition.** What is sufficient for heat production is solely the **non-bijectivity of the region map**: that several source regions fall onto a smaller target region. "Operation" is the wrong word at this point

— an operation is a description, and descriptions cost nothing. Logical irreversibility coincides with the region map only under an additional realisation assumption that has to be declared explicitly: that the logical labels are mapped onto disjoint physical regions and that no region is carried along through the transition. If that assumption is violated, logical irreversibility carries nothing — Bennett [B3] executes any logically irreversible function as a sequence of bijective region maps, with no lower bound (Section 8). And conversely, every non-bijective region map costs, whether or not it is described as an operation at all. Every physically irreversible process produces heat, regardless of whether it is described as "logically irreversible"; that is exactly the point of Section 7.3. The physics beneath the description can be far richer than the description itself.

[B] **The error of the standard presentation** is therefore this: it treats the descriptive unit — the bit — as if it were a physical minimum quantity. It is not. A physical minimum quantity does exist: h^f. But it has nothing to do with bits. The region comes first; how many bits one ascribes to it is a subsequent decision.

2 The one idea from which everything follows

A bit is physical: two **distinguishable** states of a system. Two wells for a particle, two magnetisation directions, two voltage levels. All that matters is that the system can be in state "0" or "1" and that the two can be told apart.

Region reduction: no matter which region — A or B — the system was in before, afterwards it is in the target region Z. The target region Z is freely choosable; all that counts is the ratio W_{A+B}/W_Z.

```
A   ---+
       +--->  Z
B   ---+
```

Two regions are reduced onto one — the target region Z. This map is **not invertible**: whoever knows only the target region cannot reconstruct which source region the system was in. Exactly that — and nothing else — generates the cost (Section 4).

3 What "phase space" means here

Phase space is the space of all possible microstates (all positions and momenta of all particles). A macrostate such as "region A" is not a point in it but a **region**: many microstates (thermal jitter in the left well — every jitter configuration counts as "0").

[STIPULATION] Boltzmann entropy measures this region:

$$S = k_B \ln W$$

W = phase-space volume of the region. Entropy here is a **measure of the size of the region occupied** — nothing mystical.

```
Phase space before:      Phase space after:
+-------+-------+        +-------+-------+
|       |       |        |       |       |
|   A   |   B   |   ⟶    |   Z   | empty |
|       |       |        |       |       |
+-------+-------+        +-------+-------+
 volume: 2W0 (A+B)        volume: W0 (Z only)
```

[B] Erasure halves the accessible volume:

$$\Delta S_{\text{region}} = -k_B \ln 2$$

Pure accounting over region sizes — no model, no hardware. (Generalisation to N levels and to non-uniform distributions: Section 6.)

4 Why this is not free: Liouville

The microscopic dynamics (classically Hamiltonian, quantum-mechanically unitary) has one hard property, **Liouville's theorem** [B2]:

> **[B]** The time evolution of a closed system **preserves phase-space volume**. Deforming, displacing, stirring — yes. Shrinking — never.

A refinement: Liouville holds exactly only for the closed system *without* a bath. For the subsystem "bit" alone — once coupled to the bath — it no longer holds. Only the total system bit + bath is closed; for that total the volume is preserved, while the bit subregion may shrink at the expense of the bath region.

Intuitively: the phase-space flow is an **incompressible fluid**. A blot of ink in water can be distorted arbitrarily; its volume remains.

The trap now closes:

- Erasure **demands**: $2W_0 \to W_0$ (halving, Section 3).
- Liouville **forbids**: volume shrinkage in a closed system.

[B] **Consequence: a closed system cannot erase.** Not an engineering problem — a theorem. The only way out: the system must not be closed. The volume has to go somewhere — into the bath (Section 5).

5 The bath as volume recipient

If the bit is coupled to a heat bath, Liouville applies to the **total system** bit + bath. If the bit region is to shrink by half (the uniform case: both regions equally large), the bath region must at least **double**. In the general case $\Delta S_{\text{bath}} \geq -\Delta S_{\text{bit}}$ (second law); the generalisation to $\ln(W_{\text{before}}/W_{\text{after}})$ follows in Section 6:

$$\Delta S_{\text{bath}} \geq +k_B \ln 2$$

A bath at temperature T enlarges its region by absorbing energy; the basic relation $dS = \delta Q/T$ translates this into heat:

$$\textbf{[B]} \quad Q \;\geq\; T\Delta S_{\text{bath}} \;=\; k_B T \ln 2$$

This is the Landauer bound [B1][B3]:

> **Liouville (volume is preserved) + accounting (bit region halved) ⇒ bath region doubled ⇒ heat ≥ $k_B T \ln 2$ into the bath.**

At 300 K: $k_B T \ln 2 \approx 2.87 \times 10^{-21}$ J per bit *(check script, check 1)*.

The two conditions can be read off directly:

1. **No erasure without a bath** — there is no other volume recipient (Section 4).
2. **Equality only quasistatically** — any finite speed generates additional dissipation $W \approx k_B T \ln 2 + B/\tau$ [B4][B5] (Section 10).

6 Why 2, actually? — The counting base

Binarity is **convention, not physics**. In general:

$$Q \;\geq\; k_B T \ln\left(\frac{W_{\text{before}}}{W_{\text{after}}}\right)$$

— a ratio of region sizes. N levels erased onto one: $\ln N$ *(check script, check 4)*. The $\ln 2$ is the computer-science special case.

[B] What counts as a level? Not energy. Energy levels are unsuitable as a counting base: they may be degenerate or have unequal spacings without the count changing. The criterion is **distinguishability**: two states count separately if they can in principle be told apart without error. For **quantum systems** this means orthogonality in Hilbert space; for **classical systems** it means distinguishability within the thermal resolution limit set by the bath — i.e. the separation of the states must exceed the thermal fluctuations. A bit may sit in two degenerate states of equal energy — ideally so, in fact: holding then costs nothing, only forgetting does. Level spacings act only indirectly, through the thermal occupation probabilities. The true currency is therefore the Gibbs form [B6]

$$S \;=\; -k_B \sum_i p_i \ln p_i$$

with $\ln N$ as the special case of uniform distribution. Unequally spaced, unequally occupied systems automatically carry less entropy *(check script, checks 5–6)*.

[B] The base discretisation exists — and it is h^f. Classically the phase-space volume carries units; W is defined only up to an arbitrary cell size. This does not disturb the accounting, because only differences enter — the cell cancels. Quantum mechanics fixes the cell absolutely: the number of orthogonal states in a region Γ of a system with f degrees of freedom is

$$N_{\text{max}} \;=\; \Gamma/h^f \qquad \text{[B7]}$$

(check script, check 7: colloid in a 1 µm trap, roughly 7.9×10^9 orthogonal states — the "2" of the experiment is a coarse choice, not something fundamental).

6.1 Fundamental versus emergent two-valuedness

	Spin-½	**Colloid in a double well**
origin of the 2	fundamental: Hilbert space $\mathbb{C}^2$	emergent: 2 metastable macro-regions
validity	always	only while the barrier $\gg k_B T$ on the process timescale
underneath	nothing — no finer gradation	quasi-continuum of microstates

6.2 The quasi-analogue case

A value in a range L with resolution δx carries $\ln(L/\delta x)$ levels; region reduction costs $k_B T \ln(L/\delta x)$ *(check script, check 8)*. δx is not free: the bath noise sets it — the same bath that collects the erasure cost limits the gradation. No analogue trick evades the bound.

7 Why the content is irrelevant

At **no point** did the entire calculation ask what meaning region A or B carries, whether the bit belongs to a correct computation, whether it is interpreted as "true" or "false". Only this entered: **how many** distinguishable regions before, **how many** after.

Thermodynamics counts regions; it does not read them. Region A and region B are physically equivalent — physics knows no preferred direction, and the target region Z is freely choosable. What meaning one assigns to them ("true"/"false", 0/1, high/low) is a decision outside thermodynamics:

> [B] **Region reduction costs the same for every assignment — regardless of how one interprets the regions.** There is no thermodynamic detector of content or of truth.

In the same language: a non-terminating computation dissipates energy without bound and has no truth value while doing so. Energy turnover and semantic validity lie on different levels — phase-space accounting here, representation there. A separation of categories: physical validity (hardware level) versus representational validity (content level).

7.1 Hardware and computational operation are different things

A fundamental error of reasoning in the standard presentation of Landauer: the hardware is treated as an image of the computational operation — as if the physical apparatus existed, and were thermodynamically effective, only while a logical operation is being executed. That is not so.

[B] **The hardware is not the computational operation.** A 64-bit processor is a physical system with a 64-bit state space. That state space exists independently of whether a meaningful operation is running, of whether the result requires 1 bit or 64, of what meaning the result carries, of whether anyone reads the result at all. The hardware is not an abstraction — it is silicon, wiring, fields, capacitors, interacting continuously with their environment.

[B] The computational operation is an abstraction over the hardware. It describes what the programmer or mathematician sees — input, output, function. The hardware does not know that abstraction. It knows only state transitions, charge displacements, heat release.

The consequence for Landauer. Asking "what does erasing a bit cost" takes the abstraction as the starting point and looks for the thermodynamic cost of the logical operation. That is a legitimate question — but it misses the physics of the real system. The real computer produces entropy continuously, independently of the logical abstraction currently running on it.

7.2 The same statement from the hardware side

The statement of Section 7 — thermodynamics reads no content — can be derived independently of the accounting, directly from the physical structure of the computer.

[B] The computer does not shrink. A 64-bit processor has, before and after every operation, the same number of transistors, the same number of wires, the same number of degrees of freedom. The phase space of the hardware stays constant — 2^{64} possible states before, 2^{64} after. The result may need 1 bit; the apparatus that produced it has stayed exactly as large.

[B] What actually goes into the bath is the correlation. The logical map is many-to-one: many different input states land on the same output. The 63 discarded bits are logically gone, but physically their trace still sits somewhere in the transistor states — only no longer reconstructible from the result. Exactly this non-reconstructibility, the correlation between input and output that is no longer contained in the result, goes as heat into the lattice. Not the silicon, but the forgetting of the correlation.

[B] The act of forgetting is content-neutral. A processor computing $2+2=4$ and one computing $2+2=5$ switch the same number of transistors, discard the same bit width and pay the same thermodynamic bill to the bath. Physics does not distinguish the two. A result in region A costs the same as one in region B — physics does not distinguish the regions.

This is not a new claim — it is a derivation from below. Landauer's abstraction "region reduction costs, content is irrelevant" follows from the concrete physics of the computer, without a formula. The accounting in Section 7 and the hardware structure here arrive at the same statement by different routes.

7.3 The computer always computes

A physical computer does not stop computing while it waits for a result. The clock runs, the transistors switch, the power supply delivers current — continuously, independently of whether "useful" work is happening. An idle processor computes idle loops. A waiting server sends keepalive packets. An operating system polls interrupts. There is no state "the computer is not computing" as long as it is switched on.

This means: the act of forgetting runs **always**. The thermodynamic bill — loss of correlation, heat into the lattice — does not stop, no matter whether the result is needed or not, no matter what meaning it carries.

This sharpens Landauer's formulation into practical irrelevance. He isolates a single discrete act of region reduction — "region reduction costs $k_B T \ln 2$". Reality is a **continuous current** of state transitions, losses of correlation and heat releases, three billion times per second in a modern processor, without pause.

A concrete example: dynamic RAM. A DRAM bit is a capacitor holding charge. But capacitors lose charge through leakage currents — continuously, unstoppably. Therefore every DRAM cell must be **refreshed** every 32–64 milliseconds: the refresh operation reads the state and writes it back. This happens cycle by cycle, automatically, for each of the billions of cells — no matter whether the data are currently needed, no matter what meaning they carry, no matter whether the program is asleep.

This is not a side effect — it is the physical nature of the system. The DRAM refresh is **not** an act of region reduction in Landauer's sense; it is an act of preservation, and its direct costs are primarily engineering costs (leakage currents, recharging). But the reason it has to happen at all is thermodynamic: the capacitor is permanently coupled to the heat bath. The charge — the physical region "bit charged" — is continuously pushed towards the common equilibrium by the thermal coupling. That is a permanent, unceasing interaction with the bath, independently of whether a logical operation is taking place. The refresh is the technical answer to this thermodynamic reality — it renormalises the region back, because the bath permanently blurs it.

But that is not yet the whole point. The refresh **itself** is not a passive reaction — it requires active computational work: a counter must run, the address of the next row to be refreshed must be computed, a timing signal must be generated, the memory bus must be blocked, the sense amplifier must be activated. That is real computational work — with real state transitions, real losses of correlation, a real entropy current into the bath. And for a bank of 8192 rows at a 64 ms refresh interval the memory controller issues roughly 1.3×10^5 refresh commands per second — without interruption, entirely independently of whether the running program is computing anything or asleep.

That is the actual point: it is not the refresh operation itself that is Landauer-relevant in the sense of individual bit erasures, but the fact that **the computer must compute continuously in order to maintain its own state** — refresh counters, clock generators, watchdog timers, operating-system schedulers, memory controllers. This is a permanent self-computation that has nothing to do with the user program. Landauer does not describe the thermodynamics of this self-computation, because he considered individual operations; it is an additional, more fundamental cost category: **the computer pays entropy for existing — not only for computing.**

[K] A DRAM cell that "does nothing" for an hour has nevertheless run through roughly 1.12×10^5 refresh cycles in that hour (at a 32 ms interval; at 64 ms roughly 5.6×10^4), producing entropy continuously. The stored information was the same throughout — but the thermodynamic bill kept running.

That is the physical computer as it really is: not an automaton that discretely erases bits, but a system that continuously fights thermodynamic decay and pays for it. The question is therefore not: what does region reduction cost? But: how large is the continuous entropy current of a physical computer as a function of its clock frequency, its bit width and its logical irreversibility — independently of what it happens to be computing?

That is an open question. Fredkin, Toffoli and Landauer himself thought about it [B3][B13]. The standard presentation still starts from the individual act of bit region reduction, because that is tractable. It is an abstraction that misses the physics of the real computer: the computer is not an automaton that erases individual bits on command — it is a physical system producing entropy continuously, whether one looks or not, whether the result is correct or not.

7.4 Quantum computers: decoherence as a continuous entropy current

For the quantum computer the same principle holds as for DRAM — the physical apparatus fights thermodynamic decay continuously — but the physics is more unforgiving.

What decoherence is. A qubit in superposition has a phase relation to its environment. Every smallest interaction — a phonon, a photon, a magnetic-field fluctuation — alters that phase. The qubit loses its quantum nature not because information is annihilated, but because the correlation between qubit and environment is distributed into the bath. Liouville again: the volume is preserved, but it spreads into qubit-plus-environment correlations that nobody controls any more.

The aggravation relative to DRAM. With classical DRAM one can refresh — read the state, write it back; it costs thermodynamically, and it works because 0 and 1 are robust. For a qubit this is impossible in principle: a measurement collapses the superposition. An unknown quantum state can neither be copied (no-cloning theorem [B14]) nor measured without altering it. Refresh in the classical sense does not exist.

What exists instead is **quantum error correction** [B15]. But it is thermodynamically even more expensive than DRAM refresh — it needs 100–1000 physical qubits per logical qubit, continuous syndrome measurements and classical correction operations. All of this runs on, cycle by cycle, for every logical qubit, even when no gate operation is taking place.

[K] **Decoherence time as the DRAM analogue.** A superconducting qubit typically has $T_2 \sim 100$ µs [B16] — after which the phase information is distributed in the bath. That is roughly 640 times shorter than a DRAM refresh interval of 64 ms, but qualitatively different: DRAM loses information slowly and steadily, the qubit loses it exponentially fast, and the “information” lost is not a bit but a continuous complex amplitude over a high-dimensional Hilbert space (bounded in practice by h^f, Section 6).

The entropy current in a quantum computer. For the classical computer the continuous entropy current is determined by clock frequency and bit width. For the quantum computer an additional term appears: the decoherence-induced entropy current, which runs proportional to the number of qubits and inversely proportional to T_2 — independently of whether a gate is being executed. A waiting quantum computer produces entropy more aggressively than a waiting classical computer, because the quantum nature itself fights the environment.

Decoherence is not region reduction in Landauer’s sense — it is a loss of quantum correlations. But thermodynamically it pays the same price: the correlation goes into the bath, the bath absorbs the volume. The decisive difference from DRAM: with DRAM the information can be recovered (refresh), with a qubit it cannot (no-cloning theorem [B14]). Decoherence and Landauer erasure are different processes with the same thermodynamic mechanism.

Content-neutral as always. Decoherence does not distinguish between a qubit carrying one quantum state and one carrying another. It destroys both equally fast. The thermodynamic price of quantum computing does not attach to the content — it attaches to the physics of the system.

[H] **Open question.** For the classical computer the continuous entropy current is in principle reducible by reversible logic (Section 8). For the quantum computer this is more fundamentally limited: whether there is a matter-of-principle minimum price for maintaining quantum coherence — independently of the quality of error correction — is an open question of quantum thermodynamics [B17].

8 Region operations — volume-preserving or volume-reducing

Every physical change of state is a **map between phase-space regions**. The thermodynamically decisive question is exclusively: does the map preserve the region volume, or does it reduce it?

Whether that map belongs to a "computation", whether it is described as a logical operation, whether anyone calls it meaningful — for thermodynamics this is entirely irrelevant.

[B] **Volume-preserving maps** — the region is reshaped, not reduced. Liouville applies. No thermodynamic lower bound:

```
A ---> B    (bijective: each input region onto its own output
region)
B ---> A
```

Traditionally such maps are called "reversible gates" (NOT, CNOT, Toffoli) — physically they are **bijective region maps**. Whether they are part of a computation or not is irrelevant.

[B] **Volume-reducing maps** — several input regions land on the same target region Z. The volume shrinks. The bath must absorb the difference. Thermodynamic lower bound:

```
AA ---+
AB ---+---> Z      4 input regions → 2 output regions
BA ---+             volume shrinks ⇒ the bath pays
BB -------> Z
```

Traditionally such maps are called "irreversible gates" (AND, OR, ERASE) — physically they are **non-bijective region maps**. Again: whether they are part of a computation or not is irrelevant.

[B] **Bennett** [B3] showed that any sequence of region maps can in principle be executed as a pure sequence of volume-preserving maps — intermediate states are carried along and released reversibly at the end. The thermodynamic costs do not arise at the intermediate steps but exclusively at the final relinquishing of regions:

> [B] **The lower bound does not attach to the map, but to the relinquishing of regions — to the forgetting.**

9 The accounting in one table

Step	Bit region	Bath region	Total (Liouville [B2])
before	$2W_0$	W_{bath}	$2W_0 \cdot W_{\text{bath}}$
reduction onto Z	$W_Z \leq W_0$	?	must stay $\geq 2W_0 W_{\text{bath}}$
hence	W_Z	$\geq 2\,W_{\text{bath}}$	preserved (yes)
price	$\Delta S = -k_B \ln \frac{W_b}{W_Z}$	$\Delta S \geq +k_B \ln \frac{W_b}{W_Z}$	$Q \geq k_B T \ln \frac{W_b}{W_Z}$ into the bath

10 The experiments and their limits

The image. Two glasses stand on a table — region A and region B. Next to them, the world ocean. Region reduction means: one of the glasses is poured into the ocean. The volume of the glass does not vanish — it distributes itself across 1.4 billion cubic kilometres. Liouville is satisfied: not a drop is lost.

But: which glass it was — A or B — is distributed so finely in the ocean's noise that no real instrument can ever extract it again. Not because the information were annihilated (it sits in every current, every temperature gradient), but because the resolution limit of every real observer lies ten orders of magnitude below the signal.

And: afterwards, under the same external conditions, the sea cannot offer the same representational capacity for the two glasses. It can no longer display them separately — the target region Z is smaller than $A + B$. That is not ignorance; that is a physical fact about region sizes.

The price of the pouring: $Q \geq k_B T \ln 2$. Not because of ignorance — but because the sea had to absorb the volume, and heat flowed for it.

10.1 The orders of magnitude

[K] *(check script, checks 2–3)* A real CMOS switching event lies at $\sim 10^{-17}$–10^{-16} J — four to five orders of magnitude above Landauer. All bit erasures in the world taken together would have Landauer costs in the watt to kilowatt range (depending on the assumed global erasure rate, declared in the script); in fact data centres consume some tens of gigawatts [B8] — a factor of 10^7–10^{10}. In every real machine the bound is invisible; it is a lower bound imposed by natural law, the remainder is engineering loss (CV^2 recharging, leakage currents, data transport).

[K] **The world-ocean image.** The bit volume that tips into the bath during the region reduction stands in so extreme a disproportion to the bath (1 degree of freedom against 10^{23}) that it lies **below the resolution limit** of every real instrument. The bath absorbs the volume — Liouville is satisfied — but no measuring device can still separate the signal from the bath's noise. Not because the instrument were imperfect, but because the structure of the system itself sets the resolution limit.

10.2 Epistemic versus ontological — a necessary caveat

[B] Not measurable does not mean not present. After the region reduction the microstates of the bath all still exist; every particle has a definite position and momentum;

the information about which source region (A or B) the system was in still sits in the correlations — Liouville guarantees this explicitly. The bath has absorbed the volume, not annihilated it.

Landauer does **not** hold **because** information is ontologically annihilated. It holds **because** the bath must absorb the volume (Liouville) — that is a statement about mechanics, not about knowledge or perception. A real observer cannot resolve the correlations distributed in the bath, but that is the epistemic side; the thermodynamic side (heat flow) follows from Liouville alone.

[B] What is ontologically true, however: a smaller region can carry fewer states under the same conditions than a larger one. After the region reduction $\{A, B\} \to Z$ we have $W_Z \leq W_{A+B}/2$ — the target region is at most half the size of the source region. Under the same external conditions (same temperature, same pressure, same coupling to the bath) it therefore cannot carry the same number of distinguishable states as before. This is not observer ignorance but a physical fact about region sizes. Whoever wants to represent, after the region reduction and under the same conditions, the same thing as before, needs a different — larger — physical system.

The correlations distributed in the bath are real. They are merely so finely distributed that no observer with finite means, under the same conditions, can reconstruct them. That is the transition from the ontological to the epistemic: the region is smaller (ontological), and its reconstruction would require bath-sized knowledge (epistemic).

[K] Maxwell's demon [B12] and Bennett's resolution [B3]. A being that knows all 10^{23} microstates of the bath — a Laplace demon — could trace the bit back in thought. For it, erasure would be reversible and would cost nothing. Maxwell's demon shows exactly this: an observer with complete microstate knowledge can apparently reduce entropy at no cost. Bennett's resolution of 1982 [B3]: the demon must **erase its measurement notes** in order to begin a new cycle — and exactly that costs $k_B T \ln 2$. The actual site of entropy production is the erasure of the notes, not the measurement itself. The second law is thereby preserved, but the justification is sharper: not "information is lost", but "region reduction of information produces entropy".

In summary: Landauer is grounded **ontologically** by Liouville. The **irreversibility** is statistical, not ontologically absolute — for a demon with complete knowledge it would be liftable. For every real observer, however, the resolution limit lies so far below the signal that the statistical irreversibility is practically absolute. Both must be said.

10.3 The single-particle experiments

Bérut et al. 2012 [B4] (colloid in an optical double well, cycle times 10–40 s), Jun/Gavrilov/Bechhoefer 2014 [B5] (feedback trap, higher precision). Both approach the bound and confirm the $1/\tau$ form of the additional dissipation *(check script, check 9)*. They work in the quasistatic limiting case — the smallest possible bath, extremely slow processes — in order to come anywhere near the bound at all. A real computer erases bits in a nanosecond into the world ocean, in a storm: the stirring dominates everything, the bound lies ten orders of magnitude below the noise.

Three declared points of scepticism (objections to the evidential force of the experiments, not to the principle — the principle rests on Section 4):

1. **Extrapolation:** what is confirmed is a limit ($\tau \to \infty$) that no experiment enters; the

bound is fitted as an asymptote.

2. **Signal ≈ noise:** $k_B T \ln 2$ is of the order of the thermal fluctuation of a single measurement; the result exists only as an ensemble average; some individual cycles "gain" energy.
3. **Success rate:** the protocols erase with ~ 90–95 % reliability; the accounting of failed erasures enters the balance.

[K] **Feedback extension.** With measurement and feedback the generalised second law $W_{\text{ext}} \le -\Delta F + k_B T I$ applies (Sagawa–Ueda [B9]), experimentally confirmed [B10][B11]. In the accounting, information is a resource with an energy exchange rate of $k_B T$ per nat *(check script, check 10)*.

11 Connection to open questions (internal)

- **Feedback accounting in fast systems:** computational operations at ~ns timescales lie ten orders of magnitude away from the tested quasistatic regime; there the B/τ term dominates everything. Anyone wishing to measure a Landauer-like residual in fast recurrent systems must take the Sagawa–Ueda term $k_B T I$ [B9] explicitly into the conventional budget before a residual can count as new — otherwise one is measuring feedback thermodynamics, not new physics.
- **Contact with FFGFT: none.** Landauer accounting lives at the level of statistical mechanics; the ξ scheme is untouched (its own level, its own accounting — analogous to P20/P39).

12 Summary in five sentences

Region reduction shrinks the phase-space region of the system; Liouville forbids volume shrinkage in closed systems; hence a bath must take over the volume, and the price is heat $\ge k_B T \ln(W_{\text{b}}/W_{\text{a}})$ — the $\ln 2$ only as the binary special case. Counting proceeds by distinguishability (orthogonality), not by energy; the absolute base discretisation is supplied by h^f, everything coarser is emergent counting at a chosen level. Not measurable does not mean not present: the microstates of the bath all still exist, the information about the erased bit sits in correlations — Landauer holds because of the mechanics (Liouville), not because of ignorance; the irreversibility is statistical, not ontologically absolute (Maxwell's demon, Bennett's resolution). The accounting counts regions and reads no content — every region assignment costs the same. In reality the bound is never reached, because any finite speed generates additional dissipation that dominates by orders of magnitude — the bit tips into the world ocean in a storm, and the resolution limit of the bath lies ten orders of magnitude above the Landauer bound.

13 Layer-status summary

The following table is a *selection* of the load-bearing statements; further passages are marked in the running text.

Statement	Status
Bit $\neq$ physical region (Shannon vs. Boltzmann)	[B]
$S = k_B \ln W$ as the starting point	[STIPULATION]
Region reduction $\{A, B\} \to Z$ (physics, not description)	[STIPULATION]
Region reduction $\Rightarrow \Delta S = -k_B \ln(W_{A+B}/W_Z)$	[B]
Liouville's theorem	[B]
Non-bijectivity of the region map is sufficient, not logical irreversibility	[B]
A closed system cannot erase	[B]
Landauer bound $Q \geq k_B T \ln 2$	[B]
Numerical value 2.87×10^{-21} J at 300 K	[K]
Counting base = orthogonality, not energy	[B]
h^f as the base discretisation	[B]
Gibbs form as the general entropy	[STIPULATION]
Content (true/false) does not enter the calculation	[B]
Hardware $\neq$ computational operation	[B]
The computer does not shrink	[B]
Correlation into the bath (not the silicon)	[B]
The act of forgetting is content-neutral	[B]
DRAM $\sim 1.12 \times 10^5$ refresh cycles per hour (32 ms)	[K]
No-cloning theorem	[B]
$T_2 \sim 100$ µs for superconducting qubits	[K]
Minimum price for maintaining coherence	[H]
Volume-preserving maps: no lower bound	[B]
Volume-reducing maps: the bath pays	[B]
Epistemic vs. ontological: correlations remain	[B]
Sagawa–Ueda term $k_B T I$	[K]

Balance of the table: 17× [B] (algebraic/mathematical from declared foundations) · 4× [K] (empirically tested or primary sources) · 3× [STIPULATION] (declared starting points) · 1× [H] (open research question) = 25 entries.

Appendix: Corrections relative to the working note

This A-series edition follows the working note of 24 July 2026 completely in content. The following points were changed and are declared openly here so that the changes remain checkable.

1. **State count in the colloid region unified.** The note gave 7.9×10^9 in one place and $\approx10^5$ orthogonal states in another, both citing check 7. Here 7.9×10^9 is used throughout. *To be checked:* which value the check script actually contains.
2. **Decoherence time versus refresh interval.** The note gave $T_2 \sim 100\,\mu\text{s}$ as "seven times shorter" than 64 ms; the correct factor is roughly 640. Corrected.
3. **DRAM rates separated.** The note mixed $\sim15{,}000$ refresh operations per second with $\sim112{,}000$ cycles per hour and cell; the two together are inconsistent by a factor of 10^3. Separated here: per cell roughly 1.12×10^5 cycles per hour at a 32 ms interval (at 64 ms roughly 5.6×10^4); the controller issues roughly 1.3×10^5 refresh commands per second and bank at 8192 rows and 64 ms.
4. **Layer balance corrected.** The note counted 15× [B] and 8× [K]; the table in fact contains 16× [B] and 4× [K] across 24 entries (with the additional entry from item 11, now 17× [B] across 25 entries). Recounted and corrected; at the same time declared as a *selection*, since further passages are marked in the running text.
5. **Source list completed.** [B11], [B12], [B13] and [B20] were cited in the text but not listed; entry [B17] inadvertently contained the texts of [B13] (Fredkin/Toffoli) and [B12] (Maxwell's demon). All four added, [B17] cleaned up. The reconstructed entries are marked as such in the source list and are to be checked against the working files before publication.
6. **The marker [H].** The A-series scheme knows [SETZUNG], [K], [B] and [S]. The note additionally uses [H] for "open research question". Retained here and declared in the legend — *open* whether [H] is taken into the scheme or the passage is moved to [S].
7. **Order.** The note placed section 6c before 6b. Brought into logical order here (hardware ≠ computational operation, then the hardware side, then continuous operation, then quantum computers) and carried as subsections of Section 7.
8. **Figures.** The box and arrow diagrams of the note used Unicode drawing characters; here they are set in pure ASCII so that the build stays free of missing glyphs. Content unchanged.
9. **Missing paragraph head.** The paragraph on orders of magnitude (CMOS, data centres) began in the note without a head; here it is given as the subsection "The orders of magnitude" with **[K]** .
10. **Language.** Individual grammatical errors of the German source were corrected silently.
11. **"Operation" replaced by "region map" (Landauer paragraph).** The note wrote that logical irreversibility is "a sufficient condition for heat production in a certain class

of *operations*". This contradicts the document's own programme — an operation is a description, and descriptions cost nothing. What is sufficient is solely the non-bijectivity of the *region map*. At the same time the claim is stated more sharply: logical irreversibility coincides with the region map only under a declared realisation assumption (disjoint regions, no region carried along); without it, it carries nothing, because Bennett executes any logically irreversible function bijectively. The paragraph was moreover marked **[K]** throughout, although only the Landauer quotation is a primary-source statement; the derived half now carries **[B]** . A corresponding entry has been added to the layer-status table.

Not changed: all physical statements, the layer assignments of the individual passages, the formulae, the line of argument, and the finding "contact with FFGFT: none".

Sources

[B1] R. Landauer, *Irreversibility and Heat Generation in the Computing Process*, IBM J. Res. Dev. **5**, 183 (1961).

[B2] Liouville's theorem: standard result of Hamiltonian mechanics; e.g. Landau/Lifshitz, *Mechanics*, §46; quantum-mechanical counterpart: unitarity of time evolution.

[B3] C. H. Bennett, *The Thermodynamics of Computation — a Review*, Int. J. Theor. Phys. **21**, 905 (1982); id., *Notes on Landauer's principle*, Stud. Hist. Phil. Mod. Phys. **34**, 501 (2003).

[B4] A. Bérut et al., *Experimental verification of Landauer's principle*, Nature **483**, 187 (2012).

[B5] Y. Jun, M. Gavrilov, J. Bechhoefer, *High-Precision Test of Landauer's Principle in a Feedback Trap*, Phys. Rev. Lett. **113**, 190601 (2014).

[B6] Gibbs/Shannon entropy: standard; e.g. E. T. Jaynes, *Information Theory and Statistical Mechanics*, Phys. Rev. **106**, 620 (1957).

[B7] Semiclassical state counting Γ/h^f: standard; e.g. Landau/Lifshitz, *Statistical Physics*, §7.

[B8] Data-centre energy consumption: IEA reports 2024/25, of the order of several hundred TWh per year worldwide ($\approx$ some tens of GW continuous power). The figure serves only for an order-of-magnitude comparison.

[B9] T. Sagawa, M. Ueda, *Second Law of Thermodynamics with Discrete Quantum Feedback Control*, Phys. Rev. Lett. **100**, 080403 (2008); id., *Generalized Jarzynski Equality under Nonequilibrium Feedback Control*, Phys. Rev. Lett. **104**, 090602 (2010).

[B10] S. Toyabe et al., *Experimental demonstration of information-to-energy conversion*, Nature Physics **6**, 988 (2010).

[B11] *(reconstructed)* J. V. Koski et al., *Experimental observation of the role of mutual information in the nonequilibrium dynamics of a Maxwell demon*, Phys. Rev. Lett. **113**, 030601 (2014).

[B12] *(reconstructed)* Maxwell's demon — origin of the thought experiment: J. C. Maxwell, *Theory of Heat*, Longmans (1871); standard presentation in H. S. Leff and A. F. Rex (eds.), *Maxwell's Demon 2*, IOP Publishing (2003).

[B13] *(reconstructed)* E. Fredkin and T. Toffoli, *Conservative Logic*, Int. J. Theor. Phys. **21**, 219 (1982) — reversible gates and the question of the continuous entropy current of physical computers.

[B14] W. K. Wootters and W. H. Zurek, *A single quantum cannot be cloned*, Nature **299**, 802 (1982) — no-cloning theorem.

[B15] P. W. Shor, *Scheme for reducing decoherence in quantum computer memory*, Phys. Rev. A **52**, R2493 (1995); A. M. Steane, *Error correcting codes in quantum theory*, Phys. Rev. Lett. **77**, 793 (1996) — quantum error correction.

[B16] F. Arute et al. (Google AI Quantum), *Quantum supremacy using a programmable superconducting processor*, Nature **574**, 505 (2019) — typical T_2 times of superconducting qubits ~ 100 µs.

[B17] M. Esposito and C. Van den Broeck, *Three faces of the second law: I. Master equation formulation*, Phys. Rev. E **82**, 011143 (2010); P. Kammerlander and J. Anders, *Coherence and measurement in quantum thermodynamics*, Sci. Rep. **6**, 22174 (2016) — quantum thermodynamics and the minimum price of maintaining coherence.

[B18] C. E. Shannon, *A Mathematical Theory of Communication*, Bell System Technical Journal **27**, 379 (1948) — definition of the bit as $\log_2$ of the distinguishable states; explicitly information-theoretic, not thermodynamic.

[B19] E. T. Jaynes, *Information Theory and Statistical Mechanics*, Phys. Rev. **106**, 620 (1957) — connection of Shannon entropy to Boltzmann: holds exactly only for a uniform distribution over physical microstates; not for a coarsely chosen logical resolution.

[B20] *(reconstructed)* Johnson–Nyquist noise: J. B. Johnson, *Thermal Agitation of Electricity in Conductors*, Phys. Rev. **32**, 97 (1928); H. Nyquist, *Thermal Agitation of Electric Charge in Conductors*, Phys. Rev. **32**, 110 (1928).

The entries marked (reconstructed) *were missing from the working note and have been supplied from the citation context; they are to be checked against the working files before publication.*